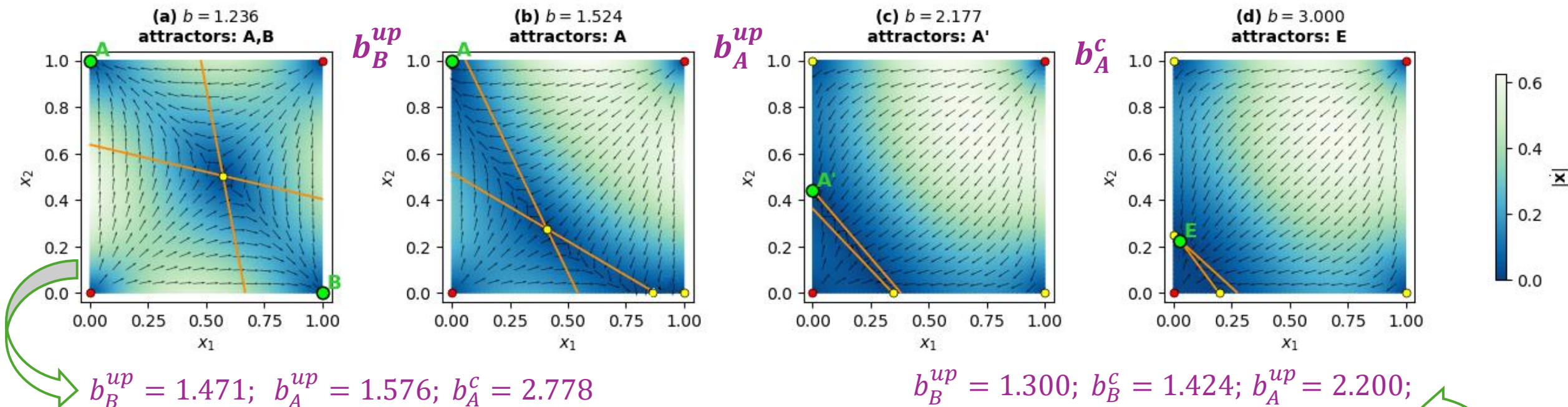
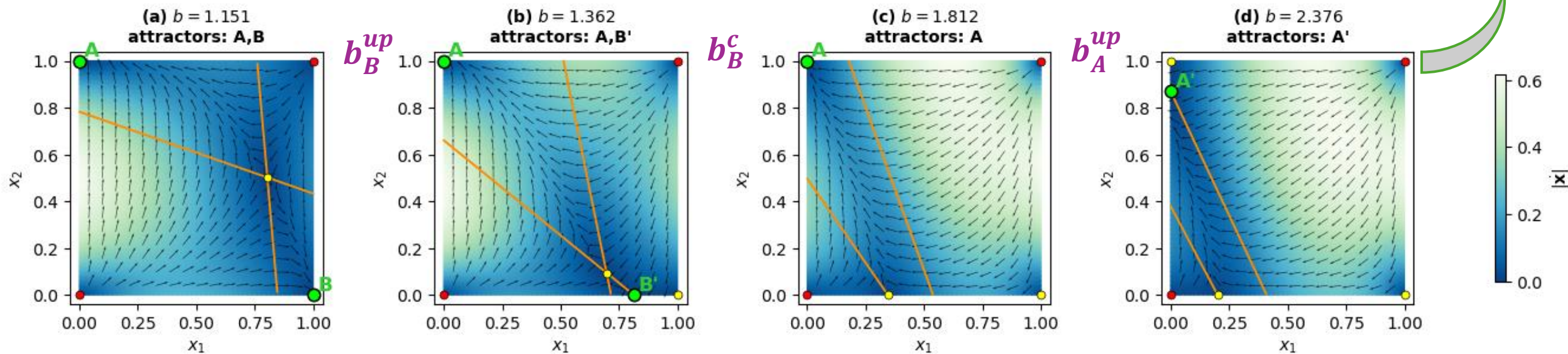


Escenario (ii₂) | $\beta = 1.105$, $r = 0.1$, $p = 0.521$, $\varepsilon = -1.0$

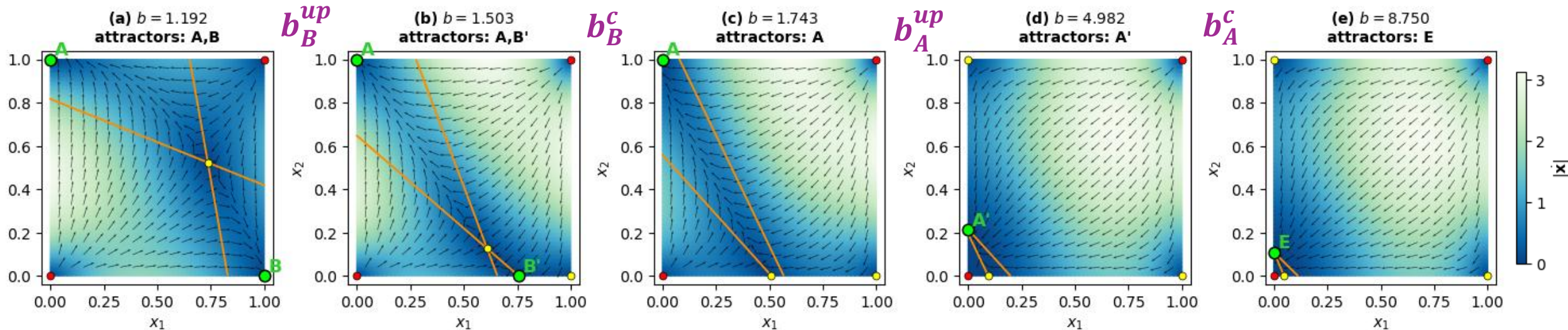


$$r_A^c = 1.200; r < r_B^c = 0.300$$

Escenario (iii₁) | $\beta = 2.0$, $r = 0.03$, $p = 0.6$, $\varepsilon = -1.0$



Escenario (iii₂) | $\beta = 1.5, r = 0.01, p = 0.575, \varepsilon = -1.0$



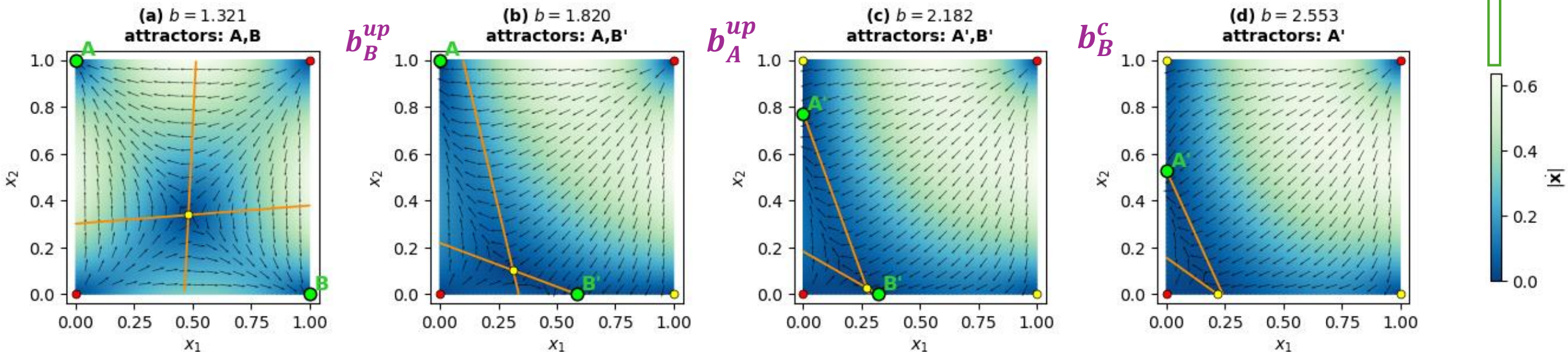
$b_B^{up} = 1.383; b_B^c = 1.623; b_A^{up} = 1.863; b_A^c = 8.102;$

$r_A^c = 0.863; r < r_B^c = 0.383$

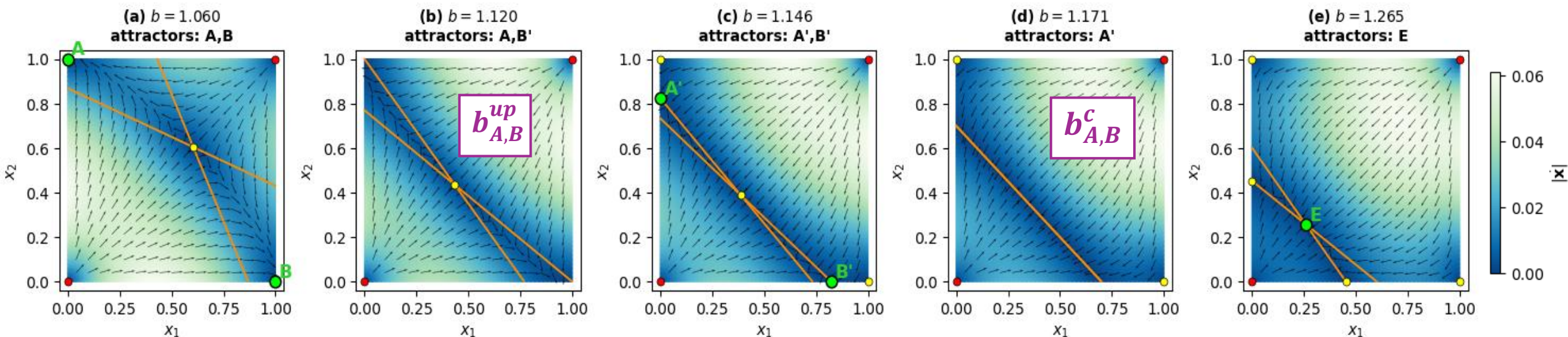
$b_B^{up} = 1.640; b_A^{up} = 2.000; b_B^c = 2.364;$

$r_A^c = 1.000; r_B^c = 0.640$

Escenario (iii₃) | $\beta = 1.25, r = 0.386, p = 0.8, \varepsilon = -1.0$

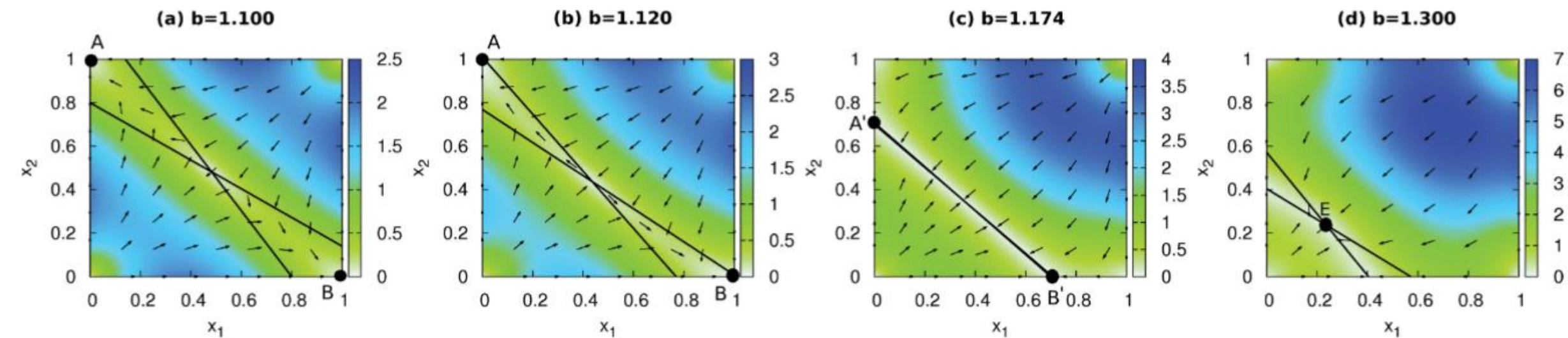


Escenario (iii₄) | $\beta = 1.0, r = 0.0, p = 0.3, \varepsilon = -0.4$



$$b_B^{up} = b_A^{up} = 1.120; \quad b_B^c = b_A^c = 1.171; \quad r < r_B^c = r_A^c = 0.120$$

—Concretamente el subcaso simétrico, denotado con s₂) en el artículo—

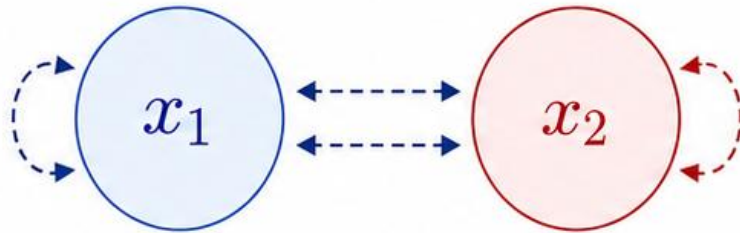


Dinámica Microscópica sobre Redes



CAMPO MEDIO

(Modelo analítico)



Ecuaciones de campo medio

$$\dot{x}_1 = F_1(x_1, x_2)$$

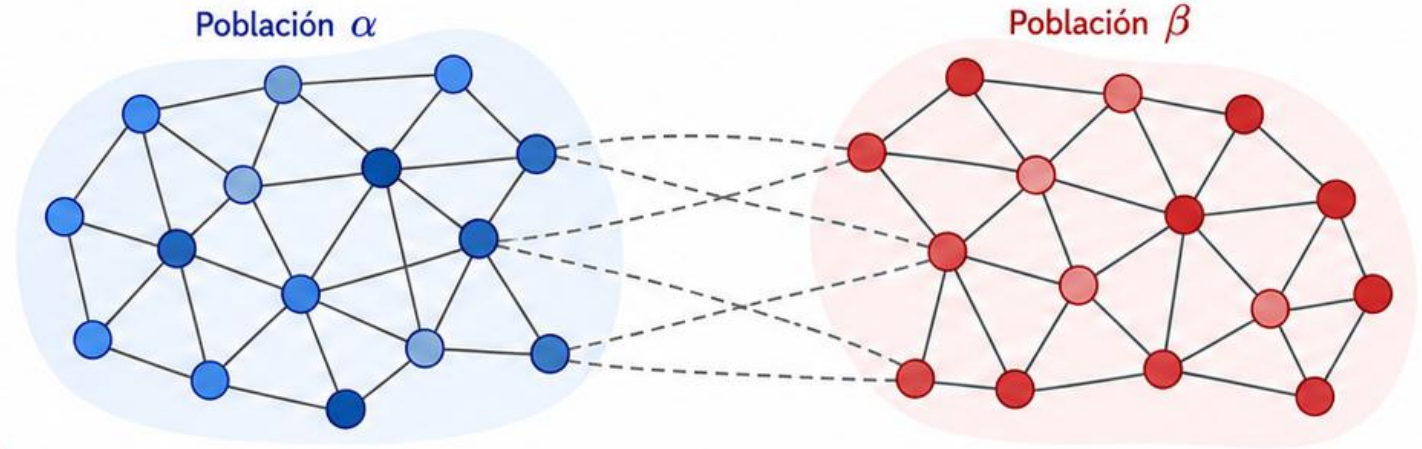
$$\dot{x}_2 = F_2(x_1, x_2)$$



Predicciones analíticas
en el límite termodinámico

MMD

(Dinámica Markoviana Microscópica)



Payoff acumulado del agente i en la población α :

$$\Pi_i^{(\alpha)}(\gamma, W) = \sum_{j \in \alpha} W_{ij}^{(\alpha, \alpha)} u_{intra}(\sigma_i, \sigma_j) + \sum_{l \in \beta} W_{il}^{(\alpha, \beta)} u_{inter}(\sigma_i, \sigma_l)$$

Probabilidad de que i adopte la estrategia de j :

$$P_{i \rightarrow j} = \mathcal{F}(\Pi_i, \Pi_j)$$

- Cooperador (C)
- Defector (D)
- Enlace intra (misma población)
- - - Enlace inter (entre poblaciones)



Interacciones locales en una red

Generación de comportamientos emergentes

Reproducción de las Figuras del Paper

Fig. 3
N1=N2=1000; k11=k22=6; x1(0)=x2(0)=0.5
reps=30; TMAX=50000

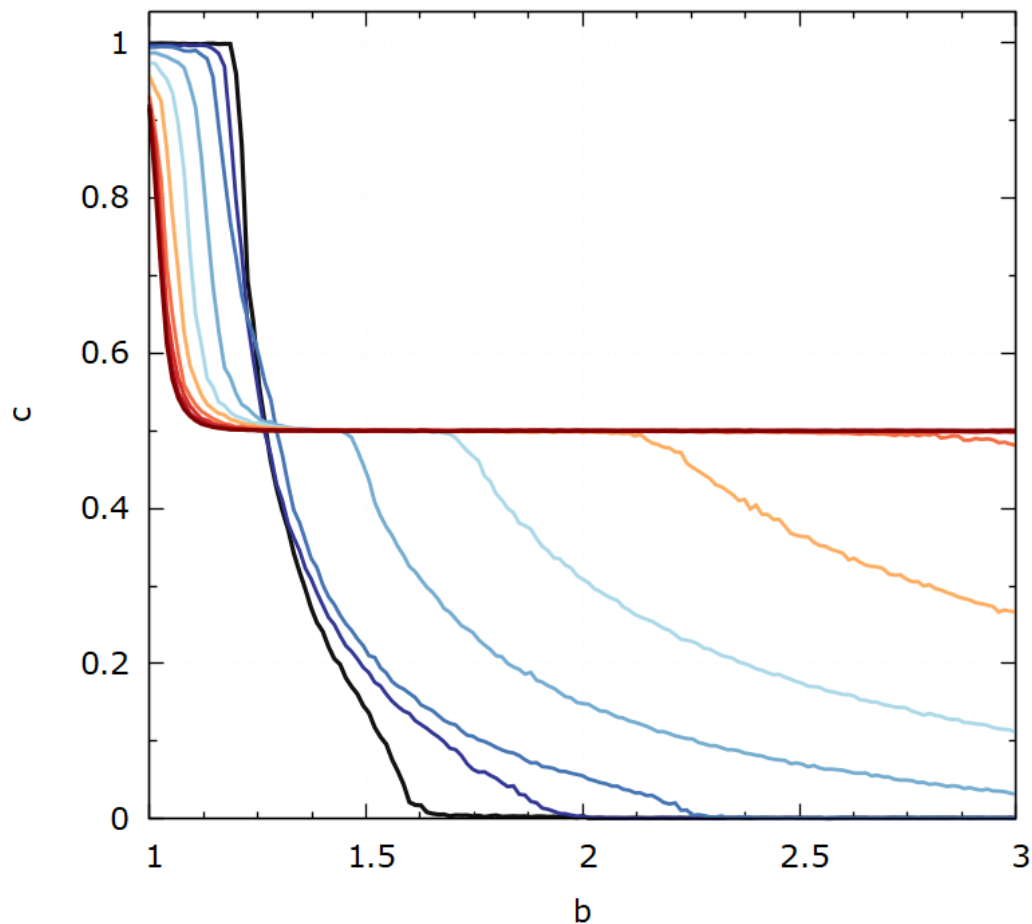
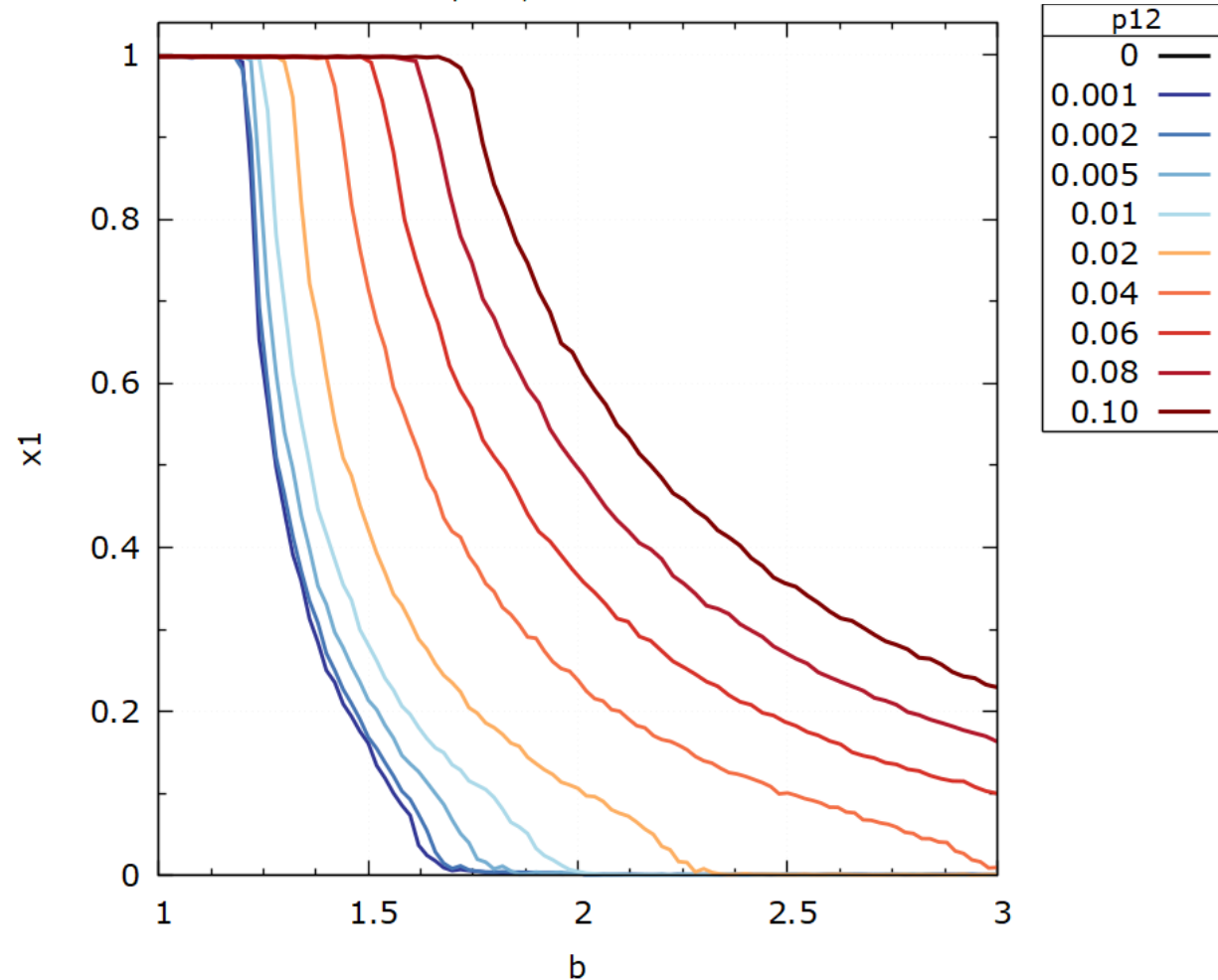
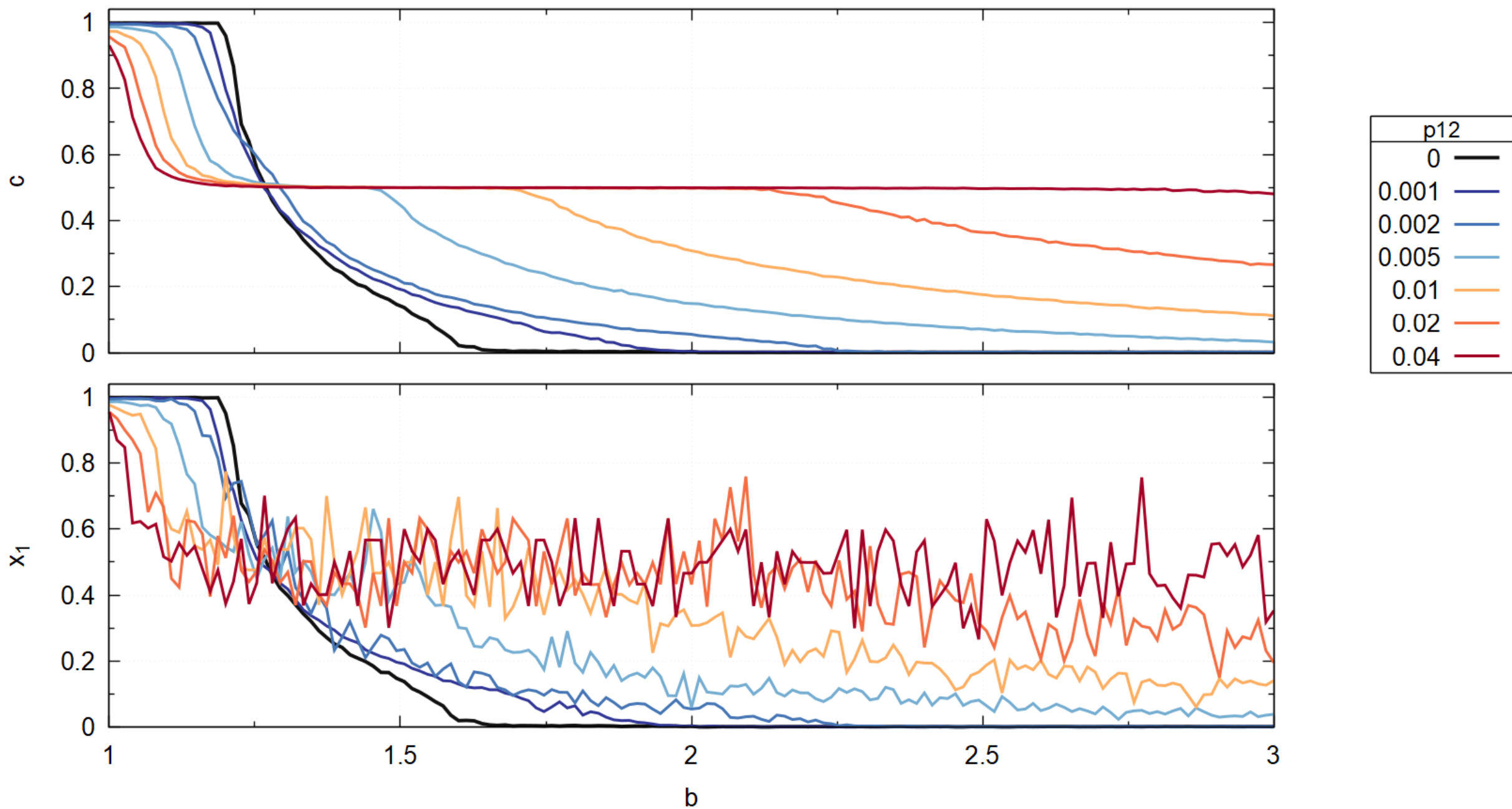


Fig. 4
N1=1000; N2=100; k11=k22=6; x1(0)=0.5; x2(0)=0
reps=30; TMAX=50000



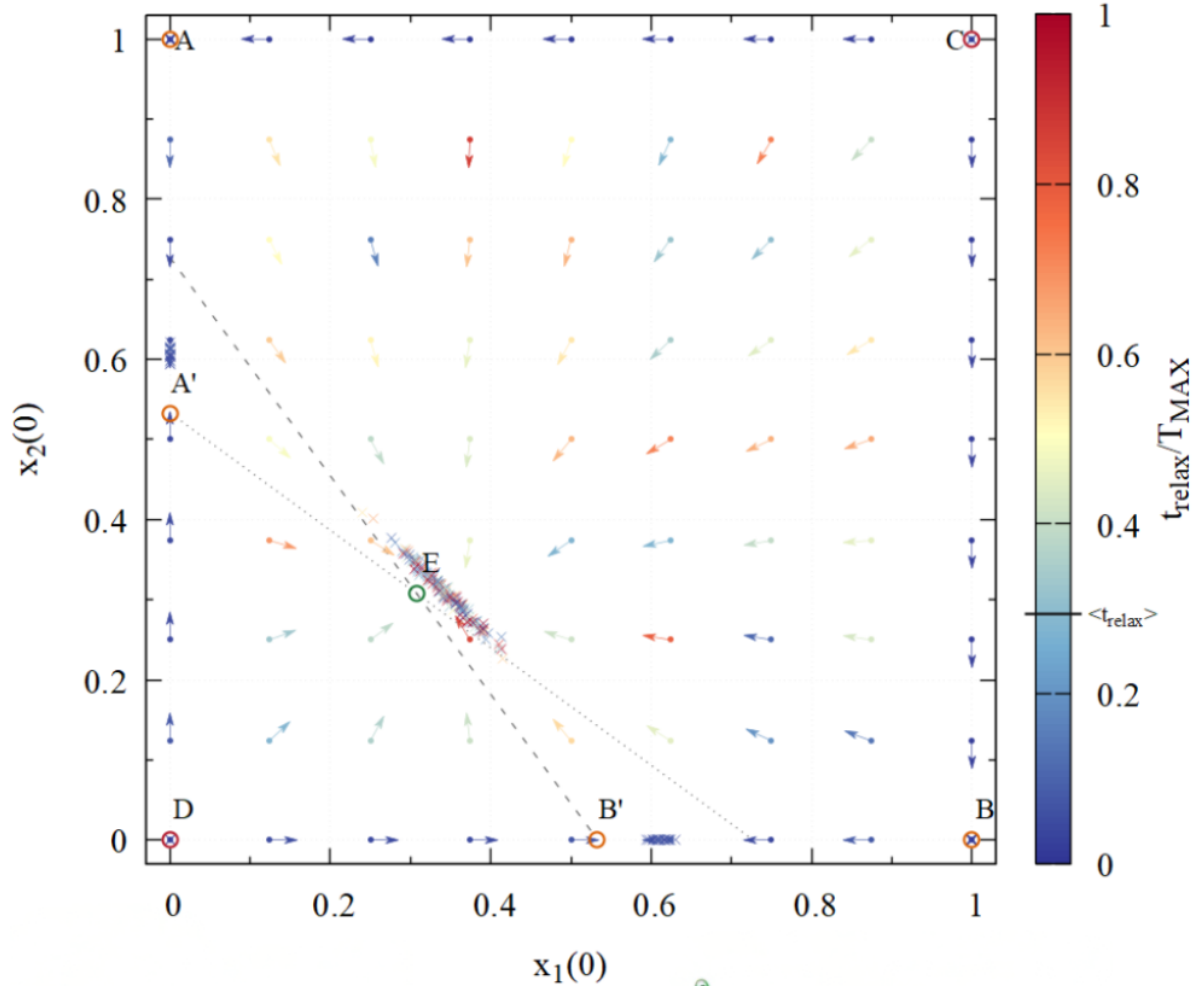
$$P_{i \rightarrow j} = \frac{\Delta_{ij}}{\max \Delta_{ij}}, \quad \Delta_{ij} \equiv \Pi_j - \Pi_i$$

Reconstrucción de la Fig. 3: cooperación global y cooperación en la red 1



Límite Termodinámico y de Alta Conectividad

$N1=500, N2=500, b=1.15, r=0, e=-0.4$
 $P11=0.80, P12=0.2, P22=0.80, ndiv=8, reps=3$



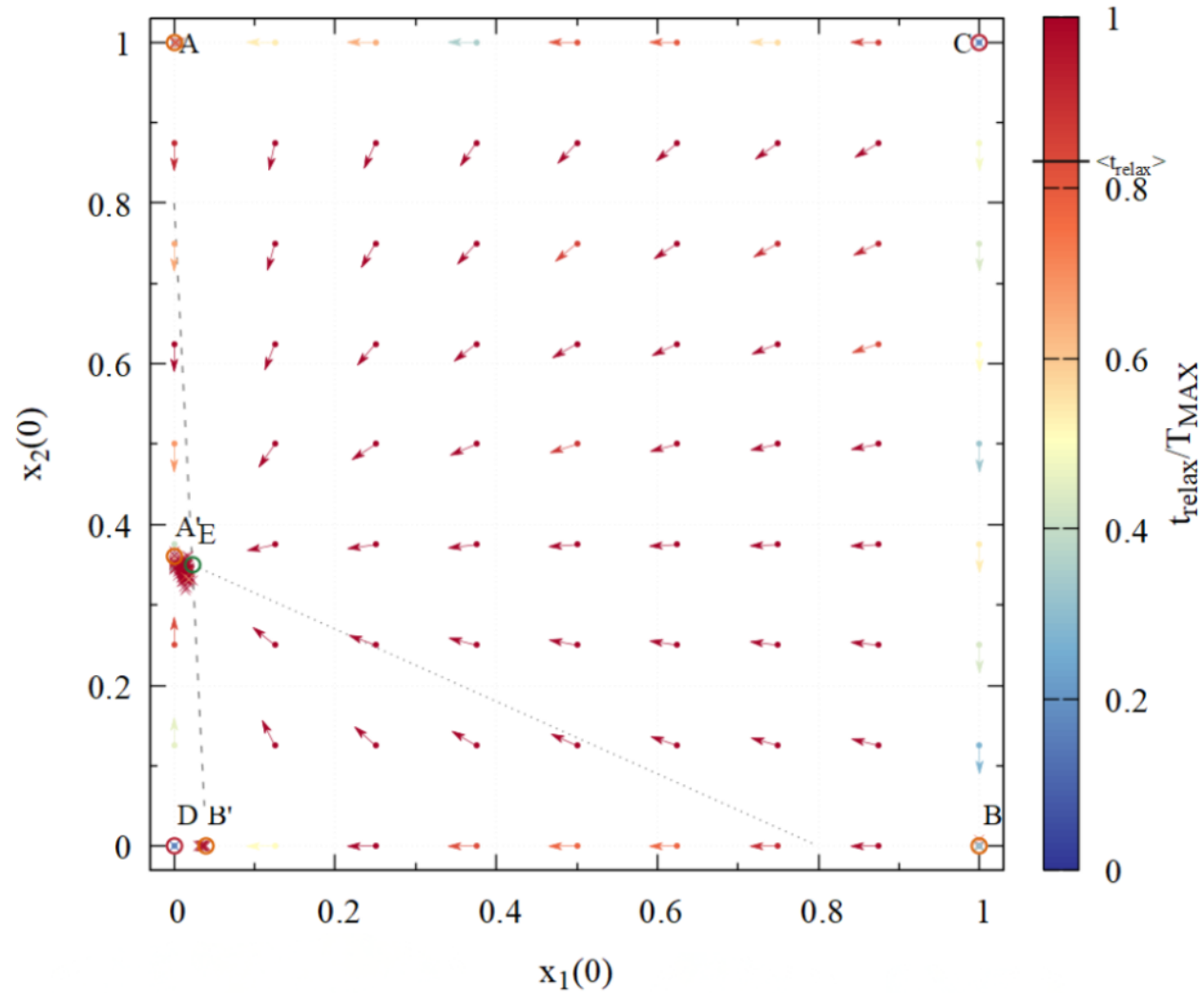
inicial, color $\propto t_{relax}$ •
 final, color $\propto t_{relax}$ x

puntos fijos:

estable
 silla

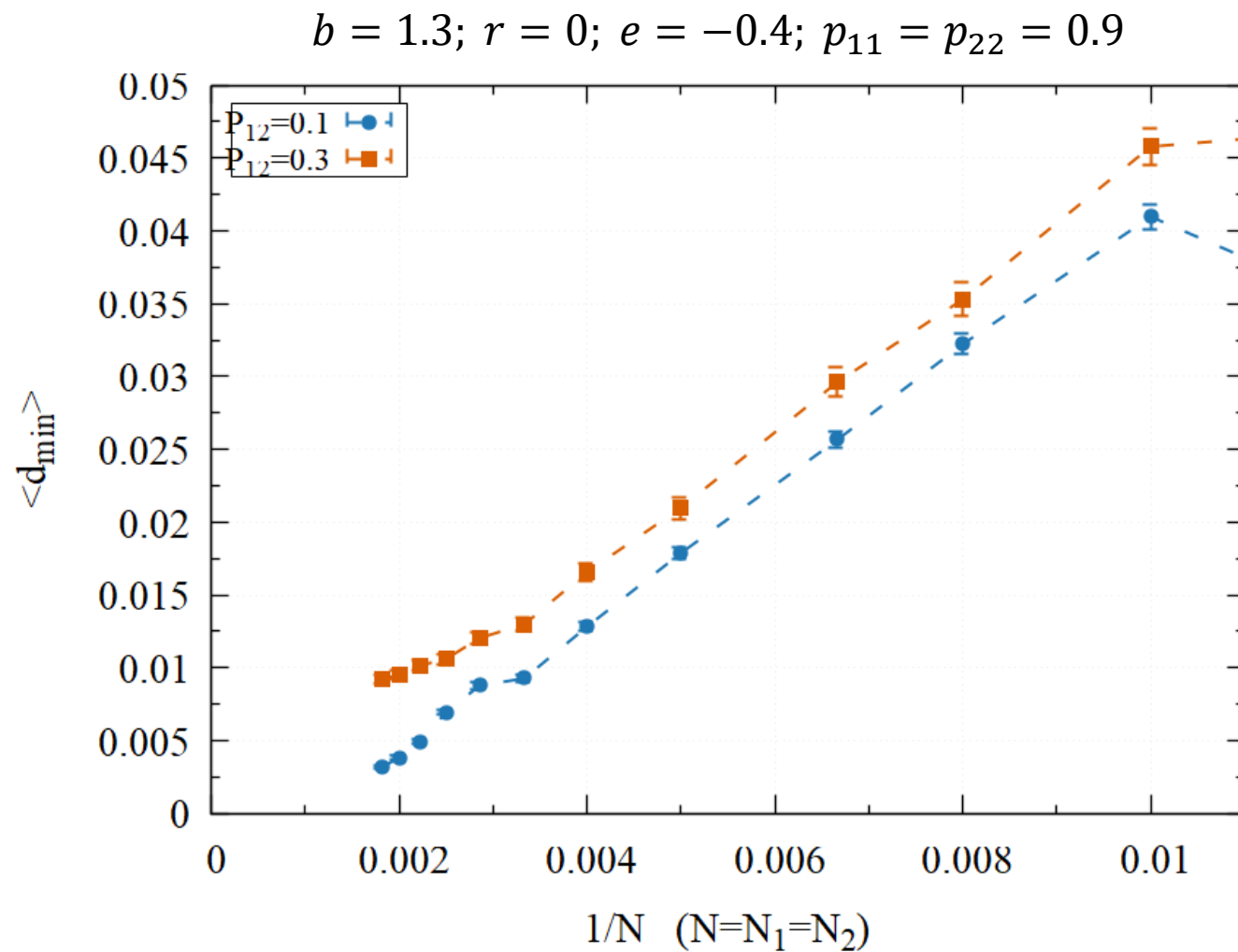
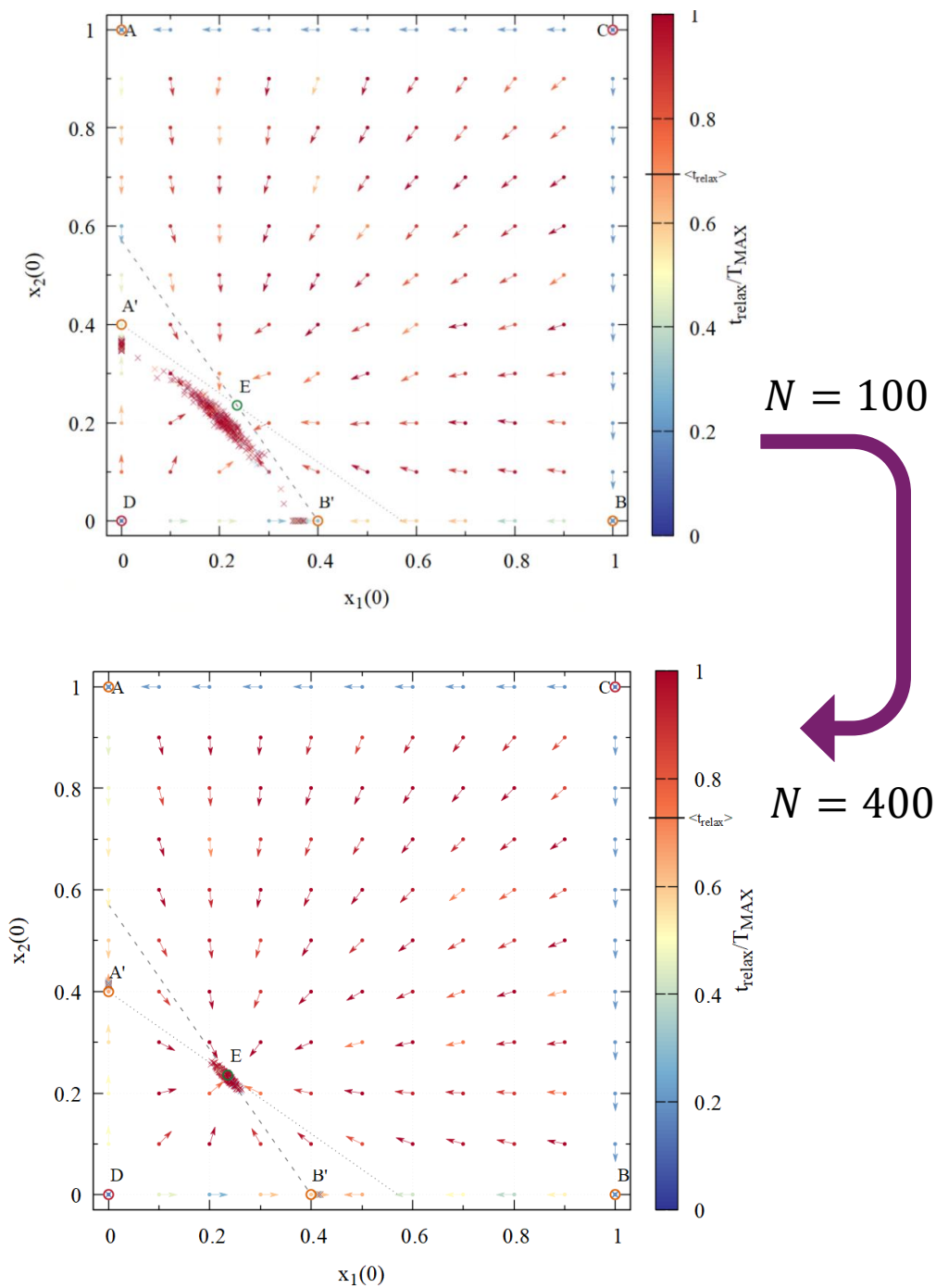
inestable
 marginal

$N1=1200, N2=400, b=1.1, r=0, e=-0.4$
 $P11=0.9, P12=0.03, P22=0.9, ndiv=8, reps=3$



1
 $\langle t_{relax} \rangle$
 0.8
 0.6
 0.4
 0.2
 0

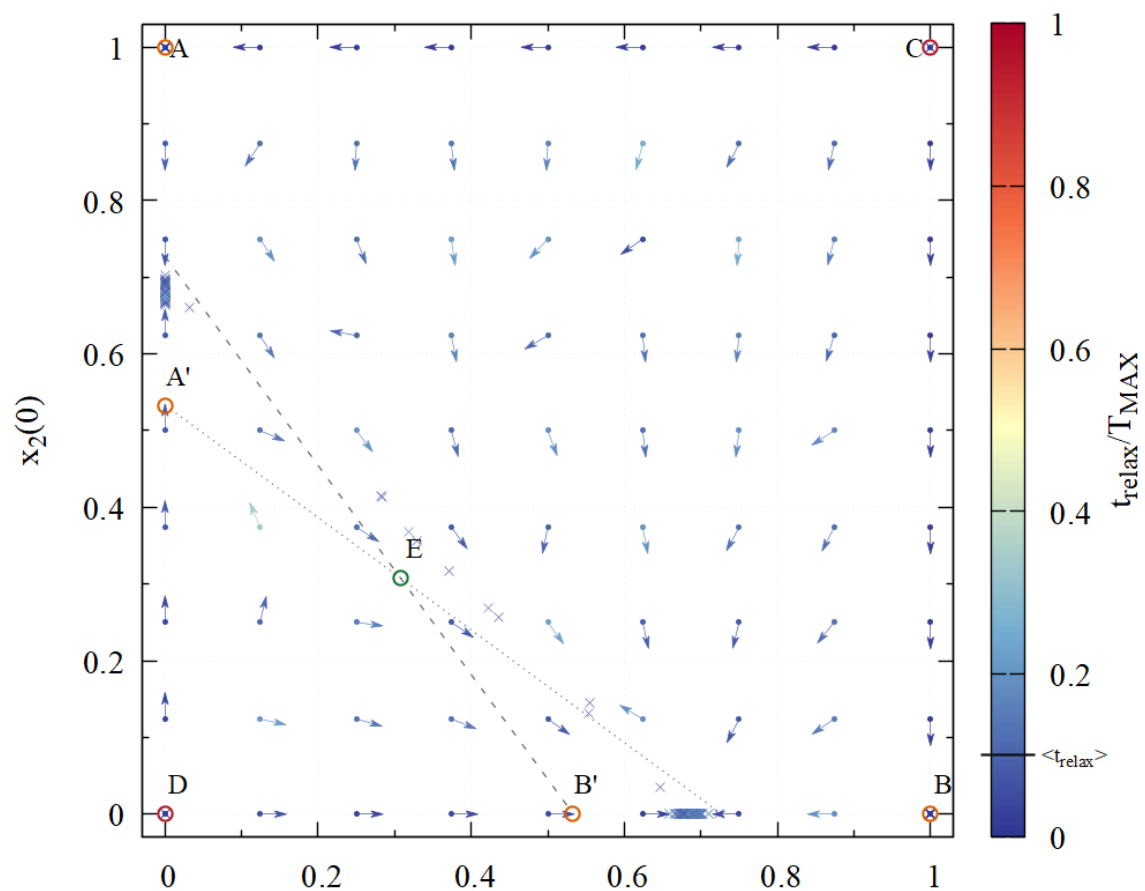
Escalado en N



Barrido en p_{11}

$N_1 = 500; N_2 = 500; b = 1.15; r = 0; e = 0.4; p_{12} = 0.2; T_{MAX} = 400000$

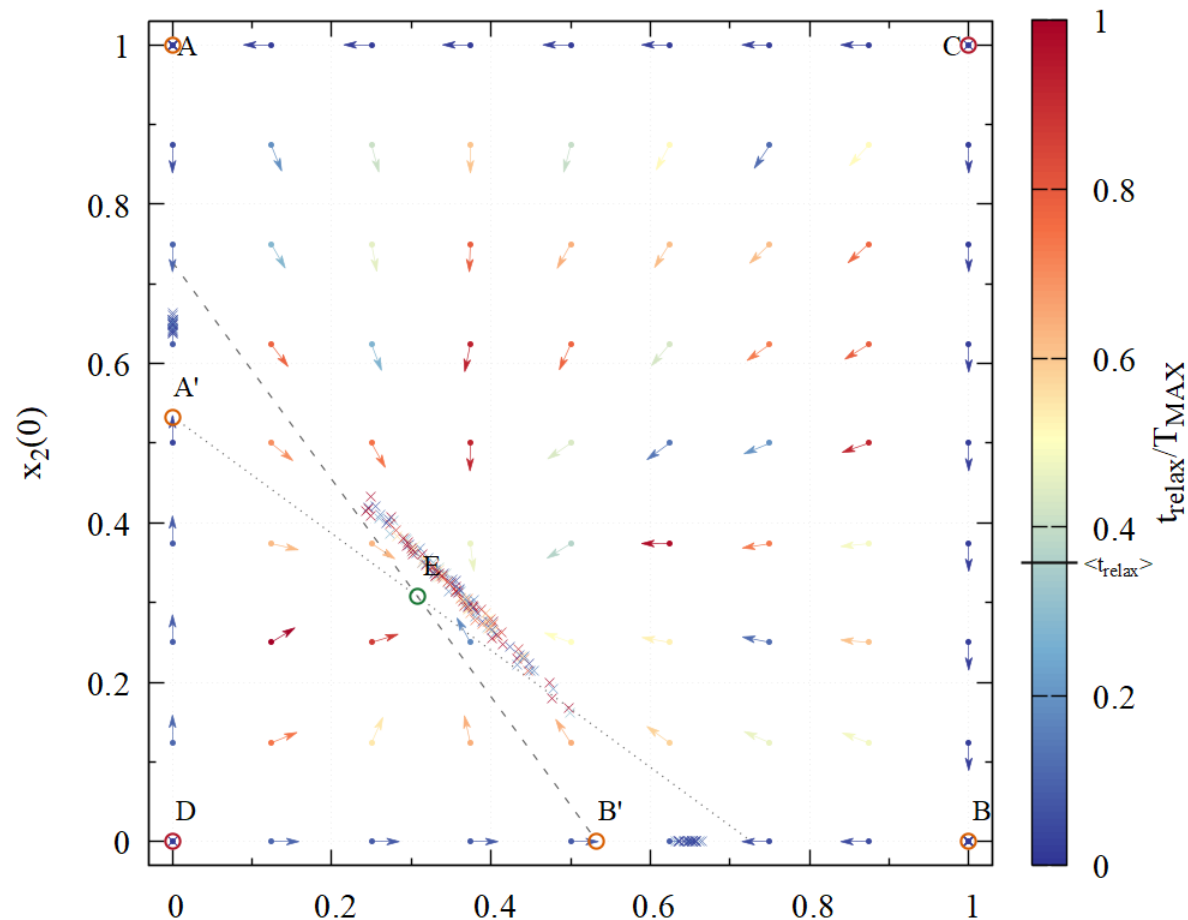
$p_{11} = p_{22} = 0.715$



inicial, color $\propto t_{relax}$ •
final, color $\propto t_{relax}$ x

puntos fijos: ● estable ● inestable
● silla ● marginal

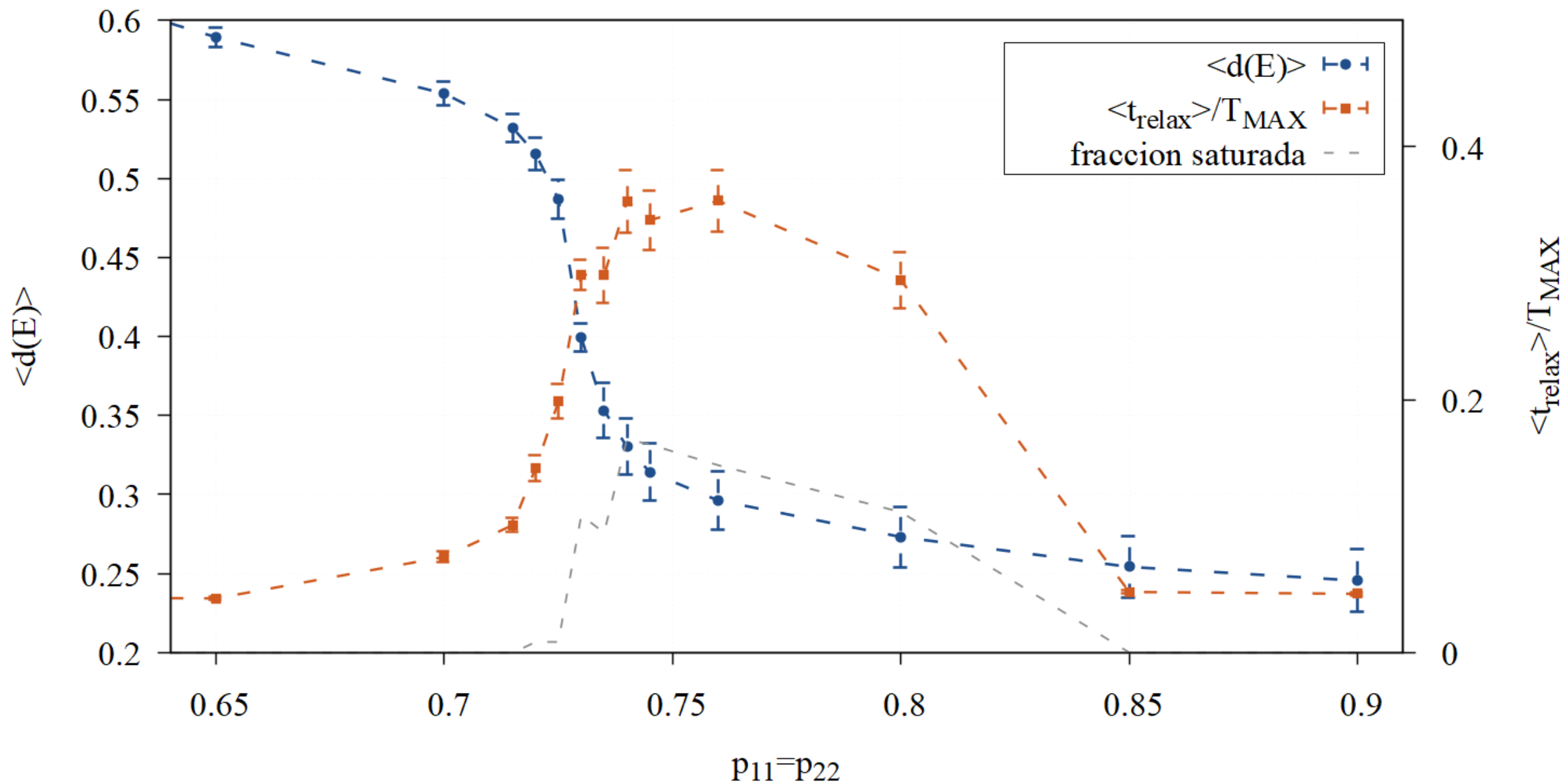
$p_{11} = p_{22} = 0.760$



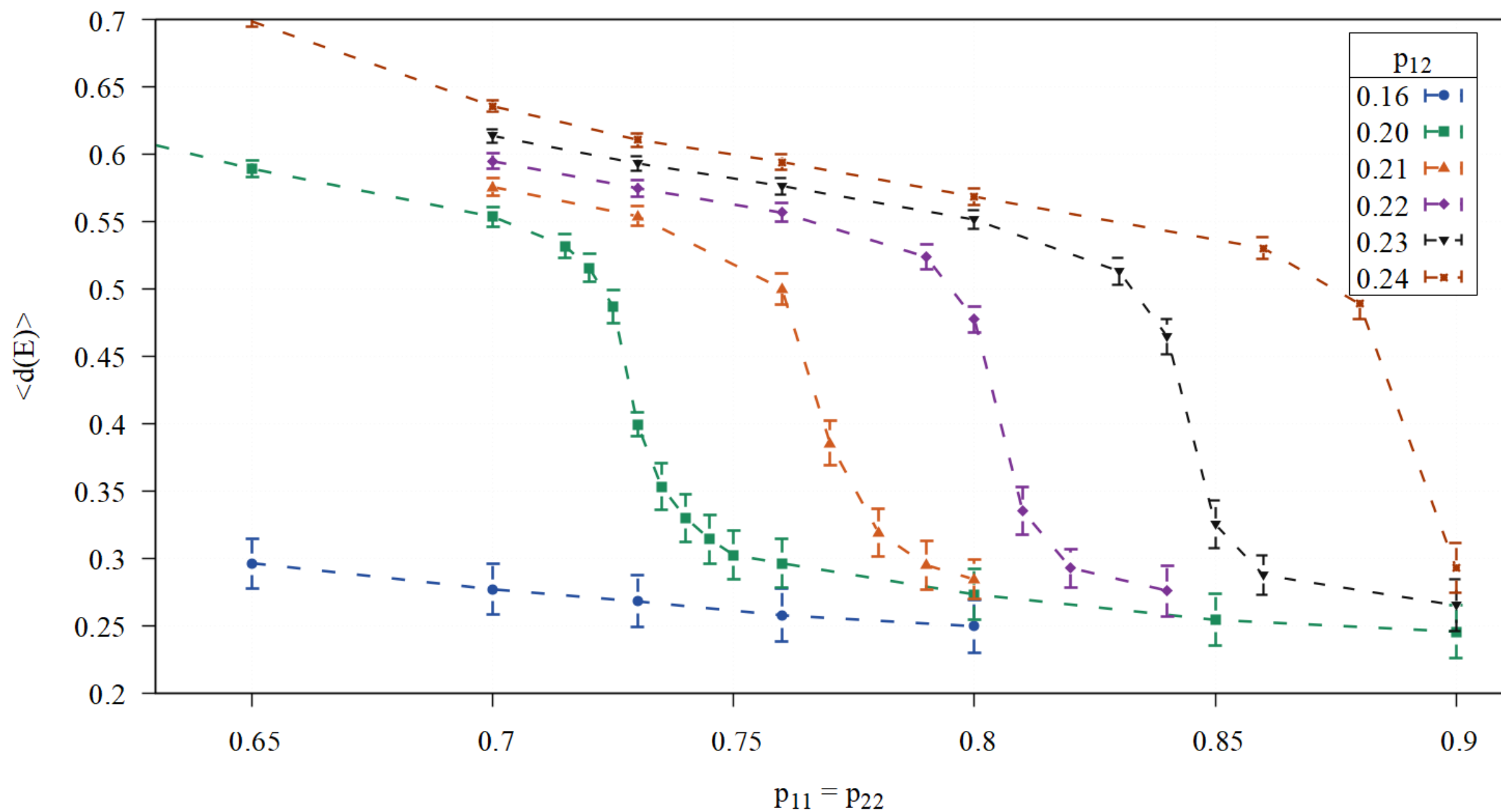
inicial, color $\propto t_{relax}$ •
final, color $\propto t_{relax}$ x

puntos fijos: ● estable ● inestable
● silla ● marginal

Transición hacia el límite analítico | $N_1=N_2=500$, $p_{12}=0.20$, $b=1.15$, $r=0$, $e=-0.4$



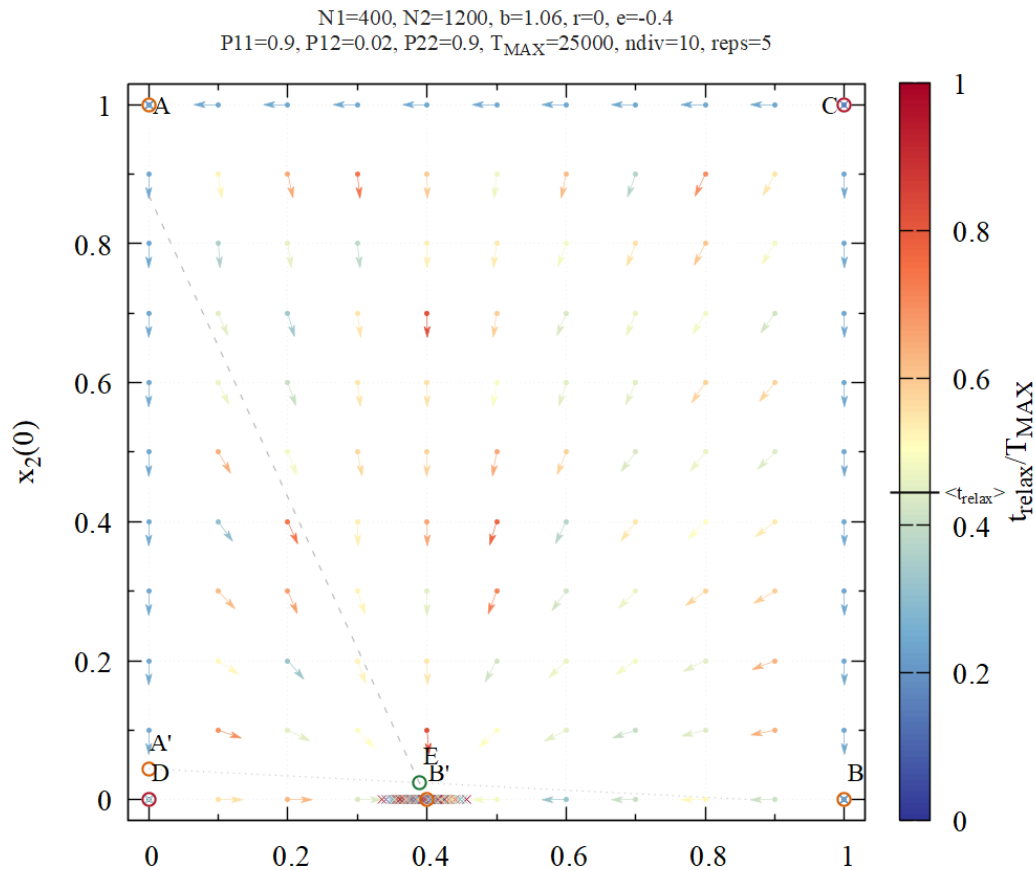
$$N_1=N_2=500, b=1.15, r=0, e=-0.4$$



Probabilidades de Copia

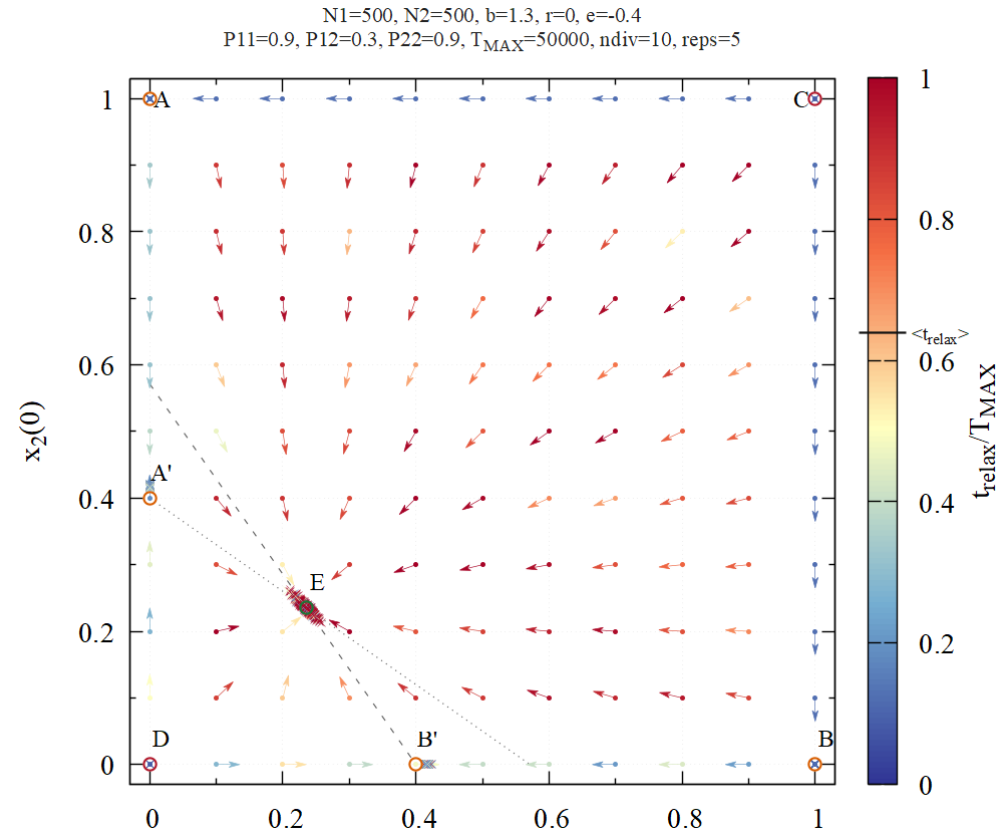
Fermi

$$P_{i \rightarrow j}^{(2)} \equiv \frac{1}{1 + \exp(-\beta \Delta_{ij} / \max \Delta_{ij})}$$



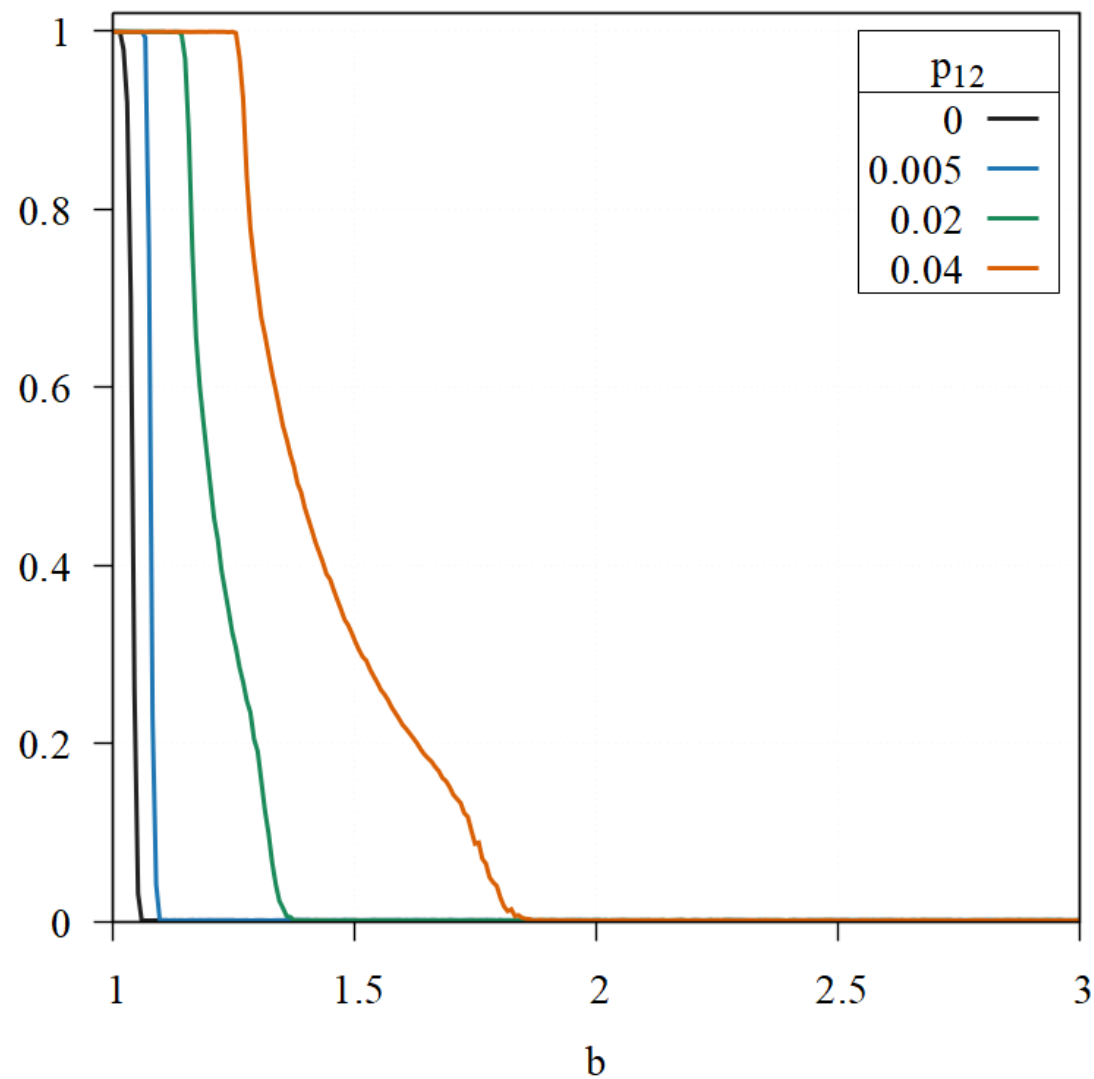
Lineal en grado

$$P_{i \rightarrow j}^{(3)} \equiv \frac{\Delta_{ij}}{\max \Delta_{ij}} \left[1 + \alpha \frac{k_j - \langle k \rangle}{\langle k \rangle} \right]$$



Probabilidades de Copia

Fermi, $\beta=5$ | $N_1=1000, N_2=100, k=6$



Lineal en grado, $\alpha=1$ | $N_1=1000, N_2=100, k=6$

